

ELEC 2FH4 - Quiz 2

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Q) The electric flux density vector inside a spherical shell of inner radius $r_i = 0.05$ m and outer radius $r_o = 0.1$ m is $\vec{D} = 3r \vec{a}_r$ $\mu\text{C}/\text{m}^2$.

(a) Evaluate the volumetric charge density inside the spherical shell.
 (b) Find the total charge inside the shell.
 (c) Calculate the total electric flux out from the shell.

- spherical shell.
- $r_i = 0.05$ m
- $r_o = 0.1$ m
- $\vec{D} = 3r \vec{a}_r$ $\mu\text{C}/\text{m}^2$

a) $\rho_v = ?$

$$\begin{aligned} \vec{\nabla} \cdot \vec{D} &= \rho_v \\ \vec{D} &= D_r(r) \vec{a}_r : \vec{\nabla} \cdot \vec{D} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 3r) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} (3r^3) \\ &= \frac{1}{r^2} (9r^2) \\ &= 9 \mu\text{C}/\text{m}^3 \end{aligned}$$

b) total charge inside shell = ?

$$\begin{aligned} Q_{\text{total}} &= \int_V \rho_v dV \\ &= \rho_v \cdot \frac{4}{3} \pi (r_o^3 - r_i^3) \\ &= 9 \cdot \frac{4}{3} \pi (0.1^3 - 0.05^3) \\ &= 9 \cdot \frac{4}{3} \pi (8.75 \times 10^{-4}) \\ &= 0.032987 \\ &= 0.0330 \\ &= 33.0 \text{ nC} \end{aligned}$$

$$V_{\text{sphere}} = \frac{4}{3} \pi r^3$$

c) total electric flux out from the shell = ?

$$\begin{aligned} \Psi &= \oint_S \vec{D} \cdot d\vec{S} = Q_{\text{enclosed by } S} \\ &= \vec{D} \cdot \vec{S} \\ &= 3r_o \vec{a}_r \cdot 4\pi r_o^2 \vec{a}_r \\ &= 12\pi r_o^3 \\ &= 12\pi (0.1)^3 \\ &= 0.037699 \\ &= 0.03770 \\ &= 37.7 \text{ nC} \end{aligned}$$

$$\vec{D} = 3r \vec{a}_r \mu\text{C}/\text{m}^2$$

$$\begin{aligned} S_{\text{sphere}} &= 4\pi r^2 \\ &= 4\pi r_o^2 \end{aligned}$$

* but $\Psi = \oint_S \vec{D} \cdot d\vec{S} = Q_{\text{enc}} = \int_V \rho_v dV$??? oops...

$$\begin{aligned} \Psi &= \oint_S \vec{D} \cdot d\vec{S} \\ Q &= \int_V \rho_v dV \\ &= \int_0^{2\pi} \int_0^\pi \int_{0.05}^{0.1} 9 \cdot r^2 \sin\theta dr d\theta d\phi \\ &\downarrow \rho_v \text{ constant} \\ Q &= \rho_v \cdot \frac{4}{3} \pi (r_o^3 - r_i^3) \end{aligned}$$

* should be $Q_{\text{total}} = \Psi$
 oops 'n'
 sorry...

c)

$$\begin{aligned} \Psi &= Q_{\text{enclosed}} = \Psi(r_o) - \Psi(r_i) \\ &= 12\pi r_o^3 - 12\pi r_i^3 \\ &= 37.7 - 4.7 \\ \Psi &= 33.0 \text{ nC} \end{aligned}$$

answer